

Sample Project Documentation

ComicCraft - AI Comic Story Creator using Gemini Models

1 Introduction

ComicCraft is a web app that turns a short story idea into a five-panel illustrated comic and PDF using Google Gemini and Stable Diffusion.

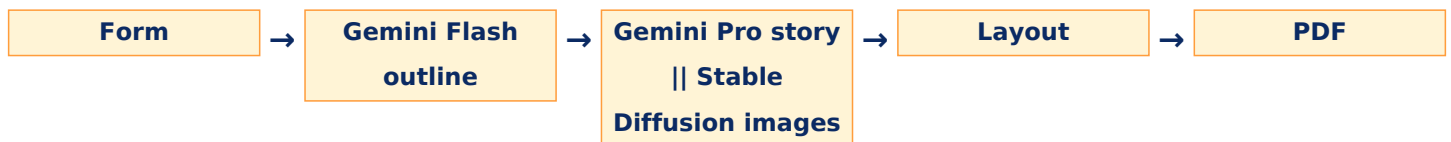
2 Problem and objectives

People without drawing skills cannot easily make comics. ComicCraft generates outline, story, images and layout automatically.

3 Requirements

- Form input: prompt, character, setting, tone, style
- Outline, story, images, preview, PDF
- JSON API and image test endpoint

4 Architecture



5 Implementation

FastAPI routes orchestrate `generate_outline`, `generate_story`, `generate_image`, `build_comic_layout` and `save_pdf`. Story and images are generated in parallel. The image pipeline is cached and warmed up at start.

6 Usage

- Start the server and open the home page
- Enter prompt and character; pick setting, tone, style
- Click Generate Comic and wait
- Review the preview; download the PDF

7 Testing

Smoke tests confirmed outline, story, images and PDF. About 12.4 s per image on Apple M5. See Performance Testing.

8 Limitations

- Gemini free quota: 20 requests per day per model
- No character consistency between panels
- CJK text unsupported in PDF
- Deprecated google-generativeai SDK

9 Future work

Speech bubbles, character consistency, hosted GPU, SDK migration, tests and Docker. See Scalability & Future Plan.